K-Traveling Repairmen Problem

***Abstract*—The K-Traveling Repairman Problem (K-TRP) is a challenging optimization problem in which multiple repairmen must serve a set of customers while minimizing the total waiting time. It extends the classical Traveling Repairman Problem by introducing multiple service agents instead of a single agent. In contrast to traditional vehicle routing problems that mainly aim to reduce travel distance, K-TRP focuses on cumulative latency, which better reflects service quality in real-world systems such as urban maintenance operations and on-demand service platforms. The problem is known to be NP-hard as computational complexity scales significantly with the number of customers and the number of service agents.**

This project introduces a two-stage solution framework for the K-TRP that combines Fuzzy C-Means (FCM) clustering with a multi-factor Fuzzy Priority Routing algorithm. Previous approaches relied on either a pure greedy algorithm, where repairmen collaboratively selected the nearest customer without spatial partitioning, or a K-Means clustering with greedy routing method, which grouped customers into compact clusters before routing. While K-Means improved efficiency compared to the greedy baseline, both methods were limited by rigid, deterministic rules that failed to account for customer ambiguity near cluster boundaries and the dynamic urgency of waiting customers.

The proposed framework overcomes these limitations by replacing hard clustering with FCM, allowing customers to hold partial memberships across clusters, and by introducing a fuzzy inference system for routing decisions. This system evaluates candidates based on distance, membership strength, and waiting time, applying eight fuzzy rules to generate priority scores. Tested on a dataset of 1,200 customers and five repairmen, the FCM + Fuzzy Priority approach reduced total latency by up to 9.69% compared to the greedy baseline and achieved more balanced workloads among repairmen. These results demonstrate that integrating soft clustering with fuzzy decision-making significantly enhances both efficiency and equity in multi-agent service routing systems.

Keywords: K-Traveling Repairmen Problem; Fuzzy Priority Routing; Greedy Algorithm; K-Means Clustering; Fuzzy Inference System; Vehicle Routing Problem; Workload Balance; Multi-Agent Routing Systems; Fuzzy C-Means Clustering; Latency Minimization.

I. Introduction

Efficient service routing and scheduling have become critical components of modern logistics, smart-city infrastructure, urban maintenance systems, and on-demand service platforms. With the rapid growth of e-commerce, field service operations, telecommunication maintenance, and emergency response systems, minimizing customer waiting time has emerged as a key performance objective. While classical routing problems such as the Traveling Salesman Problem (TSP) and Vehicle Routing Problem (VRP) primarily focus on minimizing travel distance or transportation cost, many real-world service systems are more concerned with reducing response time and improving customer satisfaction.

The Traveling Repairman Problem (TRP), also known as the Minimum Latency Problem, addresses this requirement by minimizing the cumulative waiting time experienced by customers. Unlike distance-based routing formulations, the objective of TRP is to serve customers as early as possible, thereby reducing overall latency. The K-Traveling Repairman Problem (K-TRP) extends this formulation by introducing multiple repairmen or service agents that operate simultaneously from a common depot. In K-TRP, the service sequence directly influences customer waiting times, making the problem significantly more complex than traditional routing formulations.

The K-TRP has been proven to be NP-hard, and its computational complexity increases rapidly with the number of customers and service agents. Various solution approaches have been proposed in the literature, including integer programming formulations, approximation algorithms, metaheuristic techniques, genetic algorithms, memetic algorithms, GRASP-based methods, and variable neighborhood search strategies. Although these approaches often produce high-quality solutions, they typically suffer from large computational overhead, extensive parameter tuning requirements, and limited scalability when applied to large-scale service systems.

To improve scalability, clustering-based approaches have been increasingly employed to decompose large routing problems into smaller and more manageable subproblems. Among these methods, K-Means clustering has been widely used due to its simplicity and computational efficiency. However, K-Means performs hard assignments, forcing every customer to belong exclusively to a single cluster. In practical service environments, customers located near cluster boundaries may naturally belong to multiple service regions. Such rigid assignments can lead to inefficient route construction, increased travel distance, and workload imbalance among repairmen.

Furthermore, many existing routing strategies rely on greedy nearest-neighbor selection mechanisms that consider only geographical proximity when constructing routes. While distance is an important factor, real-world service systems often require multiple decision criteria, including customer waiting time, service urgency, and regional relevance. Consequently, purely distance-based routing may fail to produce optimal latency-oriented service schedules, particularly in large and geographically

diverse service networks.

To address these limitations, this paper proposes a hybrid framework that integrates Fuzzy C-Means (FCM) clustering with a Mamdani Fuzzy Inference System for route construction. FCM introduces soft clustering by assigning membership values that represent the degree to which a customer belongs to different service regions. These membership values are subsequently incorporated into a fuzzy decision-making framework that simultaneously considers customer waiting time, cluster membership strength, and travel distance when selecting the next customer to visit. This combination enables more flexible customer assignment and more intelligent routing decisions compared to conventional clustering and greedy approaches.

The proposed methodology is evaluated on a synthetic dataset containing geographically distributed customers and multiple repairmen. Experimental results demonstrate that the integration of fuzzy clustering and fuzzy priority routing significantly reduces total latency and average travel distance compared with both Pure Greedy Routing and K-Means Clustering with Greedy Routing. The findings suggest that fuzzy decision-making provides an effective and computationally efficient mechanism for large-scale latency minimization problems.

The remainder of this paper is organized as follows. Section II presents a review of related work on the K-Traveling Repairman Problem and its variants. Section III formally defines the problem and its mathematical formulation. Section IV describes the proposed Fuzzy C-Means and Fuzzy Priority Routing framework. Section V presents the experimental results and analysis. Section VI discusses practical applications of the proposed approach. Finally, Section VII concludes the paper and outlines future research directions.

II. Literature Survey

Initially, the emphasis of the early works was mainly on the classical single depot Traveling Repairman Problem (TRP) and most of the initial theoretical work was done in terms of approximation guarantees and theoretical results. Harrelson (2003) gave one of the first results in the area of multi-agent latency minimization problems and gave an approximation scheme. Jothi and Raghavachari (2007) extended the problem and gave an approximation algorithm by incorporating repair times in the problem. However, their results deteriorate when the number of repairmen increase.

Subsequent research shifted towards incorporating more realistic constraints in the problem formulation. The multi-depot problems, such as those given in Bruni et al. (2022), used mixed-integer linear programming and hybrid genetic algorithms to solve the problem more effectively. However, these problems suffer from computational complexity when large-scale problems are considered. The profit maximization problems, such as those given in Ren et al. (2021), used time-decaying rewards and memetic algorithms to solve the problem more effectively.

Heuristic and metaheuristic techniques have also been widely adopted to improve scalability. Techniques such as GRASP combined with Variable Neighborhood Descent (Ban, 2017) and CLARANS-based clustering approaches (Hmayer and Ezzine, 2013) breaks down the global K-TRP into smaller subproblems. While these techniques reduce the computational complexity, they often require extensive tuning of parameters and may not necessarily provide an efficient workload balance among repairmen.

Dynamic and stochastic extensions of the Traveling Repairman Problem further expanded the scope of the problem. Lu et al. proposed asymptotically optimal algorithms for dynamic variants, partitioning service regions into subareas. However, region partitioning increases structural complexity and may introduce boundary inefficiencies. Similarly, works integrating machine learning or probabilistic failure modeling, such as Tulabandhula et al. (2011), attempt to incorporate predictive information but rely heavily on historical data availability.

Despite these advancements in TRP, it is apparent that there are significant gaps in literature. First, many approaches emphasize either strong theoretical guarantees or computational heuristics, but rarely balance both simplicity and scalability. Second, spatial clustering is often treated as a preprocessing step rather than being systematically integrated into latency minimization. Third, workload balance among repairmen is frequently overlooked, leading to uneven service distributions. Finally, several models depend on complex optimization frameworks that may not be suitable for real-time or large-scale smart city applications.

The literature review indicates that existing K-TRP solution approaches primarily focus on either optimization quality or computational efficiency, often sacrificing one objective for the other. In addition, the majority of clustering-based methods employ hard partitioning techniques that are unable to effectively handle boundary customers and overlapping service regions. Similarly, conventional routing heuristics frequently rely on distance-based decision rules and overlook other important factors such as customer waiting time and service-region relevance. These observations highlight the need for a routing framework capable of handling uncertainty while remaining scalable for large problem instances. Motivated by these challenges, this work proposes a hybrid Fuzzy C-Means and Fuzzy Priority Routing framework that combines soft customer assignment with fuzzy decision-making to improve latency minimization, workload balancing, and overall service efficiency.

TABLE I: Summary of Major Research Efforts on Variants of the K-Traveling Repairman Problem (K-TRP)

Title	Problem	Approach	Key Contribution	Limitation
Bruni et al. (2022)	Multi-Depot K-TRP	MILP + Hybrid GA	Efficient routing	Inefficient for large datasets
Tulabandhula et al. (2011)	TRP with Failure Probabilities	ML + TRP	Integrates predictive failure probabilities	Requires historical data
Jothi & Raghavachari (2007)	K-TRP with Repair Times	Approximation algorithms	Provable approximations	Weak for large k
Ren et al. (2021)	Multi-Repairman TRP with Profits	Memetic Algorithm	Handles time-decaying profits	Focuses on profit optimization
Harrelson (2003)	Classical K-TRP	MST-based approximation	8.497α-approximation guarantee	Primarily theoretical
Ban (2017)	Distance Constraints	GRASP + VND	Robust heuristic framework	Computationally intensive
Samuel Nucamendi-Guillén et al.	K-TRP	Mixed Integer Formulation	Provides optimal solutions	High computational cost
Amira Hmayer & Imen Ome Ezzine (2013)	CLARANS-based K-TRP	CLARANS heuristic	Decomposes K-TRP into subproblems	Limited scalability
Avrim Blum et al.	Minimum Latency Problem	Constant-factor approximation	Theoretical latency bounds	Weak practical scalability
Xiangwen Lu et al.	Dynamic TRP	Asymptotically optimal algorithm	Region partitioning strategy	Increased structural complexity

III. Problem Statement

K Traveling Repairman Problem (K-TRP), is a latency version of the Traveling Salesman Problem (TSP). Instead of one repairman, k repairmen (vehicles/agents) start from one or more depots and must serve all customers. The objective is to minimize the total latency (sum of waiting times of all customers) rather than just total travel distance.

In this problem, multiple repairmen (or service vehicles) depart from a common depot and must collectively serve a set of geographically distributed customers. Each customer requires exactly one visit and service is completed upon arrival. The performance measure of interest is total latency, defined as the sum of arrival times at all customer locations. This objective reflects service quality, as earlier visits reduce overall customer waiting time.

A. Basic Assumptions

- Single depot - All repairmen originate from a common starting point(depot).
- Homogeneous vehicles - All vehicles operate with identical speeds and have no capacity constraints.
- Deterministic travel costs - Travel times between all pairs of nodes are fixed and known in advance.
- Latency-based objective - The cost associated with each customer equals the time elapsed from departure until the repairman reaches that customer.

To visually illustrate the structure of the K-Traveling Repairman Problem, consider the example network shown in Figure 1. The network consists of a single depot (node 0) and multiple customer nodes spatially distributed across a service region. Each customer must be visited exactly once by one of the repairmen. The bold solid and dotted paths represent two feasible routes assigned to different repairmen. The objective of K-TRP is not merely to minimize total travel distance, but to minimize total latency, defined as the cumulative waiting time of customers along each route.

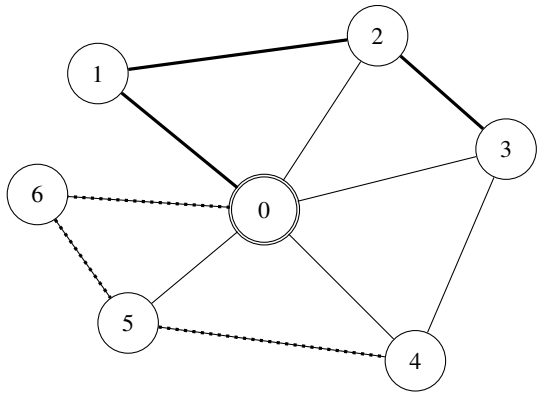


Fig. 1: Illustrative K-Traveling Repairman Problem instance with a randomly distributed customer layout. Bold solid and thick dotted paths represent routes assigned to different repairmen.

In this context, latency accumulates as a repairman progress along a route. Customers visited later experience larger waiting times due to cumulative travel distance. Therefore, route design directly influences total system latency. This illustrates the fundamental difference between K-TRP and classical distance-based vehicle routing

formulations.

B. Mathematical Formulation

Let $V = \{0, 1, 2, \ldots, n\}$ denote the set of nodes, where 0 represents the depot.

$$\min \sum_{i=1}^{n} L_i$$

Constraints:

1) k denotes the no of repairmen.
2) d_{ij} denotes the travel time between nodes i and j.
3) Flow conservation - Route continuity is maintained for each repairmen.
4) If a repairmen visits customer j immediately after customer i, then the latency at node j must satisfy $L_j \geq L_i + d_{ij}$

This objective ensures that customers are served as early as possible, reducing overall waiting time and the formulation captures the essence of latency minimization in multi-agent service systems and forms the mathematical foundation for developing heuristic or clustering-based solution methods.

IV. Proposed Methodology

To address the computational challenges associated with the K-Traveling Repairman Problem (K-TRP), a hybrid framework combining Fuzzy C-Means (FCM) clustering and Mamdani Fuzzy Priority Routing is proposed. Instead of solving the entire latency minimization problem globally, the approach decomposes the customer set into multiple service regions and subsequently performs intelligent route construction within each region.

The proposed methodology consists of four major stages:

1) Feature Extraction and Normalization
2) Fuzzy C-Means Clustering
3) Repairman Assignment
4) Mamdani Fuzzy Priority Routing

A. Stage 1: Feature Representation and Normalization

Let

$$C = \{c_1, c_2, \ldots, c_n\}$$

represent the set of customers.

Each customer is represented using a feature vector consisting of spatial and categorical attributes:

$$x_i = (Location_X, Location_Y, Distance_{Depot}, PlaceType)$$

where:

- $Location_X$ and $Location_Y$ denote customer coordinates.
- $Distance_{Depot}$ denotes the Euclidean distance from the depot.
- $PlaceType$ represents the category of service location.

Since these attributes exist on different numerical scales, feature normalization is performed prior to clustering. Standard score normalization is employed:

$$z_i = \frac{x_i - \mu}{\sigma}$$

where μ denotes the feature mean and σ represents the feature standard deviation.

Normalization ensures that no single feature dominates the clustering process.

B. Stage 2: Fuzzy C-Means Clustering

Traditional clustering methods such as K-Means assign each customer exclusively to a single cluster. However, customers located near cluster boundaries may naturally belong to multiple service regions.

To address this limitation, Fuzzy C-Means (FCM) clustering is employed.

Let

$$U = [\mu_{ij}]$$

represent the membership matrix, where

$$0 \leq \mu_{ij} \leq 1$$

denotes the degree of membership of customer i in cluster j.

The membership constraints are defined as:

$$\sum_{j=1}^{K} \mu_{ij} = 1$$

for all customers.

FCM minimizes the objective function:

$$J_m = \sum_{i=1}^{n} \sum_{j=1}^{K} \mu_{ij}^m \|x_i - v_j\|^2$$

where:

- m denotes the fuzzification coefficient.
- v_j represents the centroid of cluster j.
- $\|x_i - v_j\|$ denotes the Euclidean distance.

Cluster centroids are updated iteratively using:

$$v_j = \frac{\sum_{i=1}^{n} \mu_{ij}^m x_i}{\sum_{i=1}^{n} \mu_{ij}^m}$$

The optimization process continues until convergence.

Unlike hard clustering methods, FCM creates soft cluster boundaries that allow customers to partially belong to multiple service regions.

C. Stage 3: Repairman Assignment

Following clustering, each cluster is assigned to a dedicated repairman.

The optimal number of clusters is determined using the Elbow Method based on the Within Cluster Sum of Squares (WCSS):

$$WCSS = \sum_{j=1}^{K} \sum_{x_i \in C_j} \|x_i - v_j\|^2$$

Each customer is assigned to the cluster corresponding to its highest membership value.

As a result, the original K-TRP is decomposed into K smaller routing problems, significantly reducing the search space and improving scalability.

D. Stage 4: Mamdani Fuzzy Priority Routing

After cluster formation, route construction is performed independently within each cluster.

Unlike conventional nearest-neighbor routing, which relies solely on distance, the proposed method employs a Mamdani Fuzzy Inference System (FIS) that evaluates multiple decision factors simultaneously.

Three fuzzy input variables are considered:

1) Distance: Distance between the repairman and customer.

Linguistic terms:

- Near
- Medium
- Far

2) Waiting Time: Accumulated waiting time experienced by the customer.

Linguistic terms:

- Low
- Moderate
- High

3) Membership Strength: FCM membership value associated with the customer.

Linguistic terms:

- Weak
- Moderate
- Strong

The output variable is:

Service Priority

- Low
- Medium
- High

Triangular membership functions are employed for all fuzzy variables.

4) Fuzzy Rule Evaluation: Routing decisions are generated using a collection of Mamdani IF–THEN rules.

Examples include:

1) IF Distance is Near AND Waiting Time is High THEN Priority is High.
2) IF Membership Strength is Strong AND Distance is Near THEN Priority is High.
3) IF Distance is Far AND Waiting Time is Low THEN Priority is Low.
4) IF Membership Strength is Weak AND Distance is Far THEN Priority is Low.

The fuzzy inference engine evaluates all activated rules and aggregates the resulting fuzzy output sets.

5) Defuzzification: To obtain a crisp priority value, centroid defuzzification is performed.

The final priority score is calculated as:

$$Priority^* = \frac{\int z\mu(z)\,dz}{\int \mu(z)\,dz}$$

where:

- $\mu(z)$ denotes the aggregated membership function.
- $Priority^*$ represents the final crisp priority score.

The next customer to be visited is selected as:

$$u^* = \arg\max(Priority_i)$$

This process is repeated until all customers within the cluster have been serviced.

6) Latency Computation: As the repairman traverses the route, cumulative travel distance is continuously updated.

Let

$$T_i$$

represent the arrival time at customer i.

The latency associated with cluster j is defined as:

$$L_j = \sum_{i=1}^{|C_j|} T_i$$

The total system latency is then computed as:

$$TotalLatency = \sum_{j=1}^{K} L_j$$

Minimizing this objective ensures that customers receive service as early as possible, thereby reducing cumulative waiting time.

7) Computational Perspective: The computational complexity of the proposed framework consists of two major components.

The complexity of FCM clustering is:

$$O(nKI)$$

where:

- n = number of customers,
- K = number of clusters,
- I = number of FCM iterations.

The routing phase has complexity:

$$O\left(\frac{n^2}{K}\right)$$

Although fuzzy inference introduces additional computations, the decomposition of the original K-TRP into smaller cluster-specific routing problems significantly improves scalability.

Consequently, the proposed FCM + Fuzzy Priority Routing framework remains computationally efficient while achieving superior latency reduction compared to traditional clustering and greedy routing approaches. "'

E. Algorithm

Algorithm 1 FCM Clustering + Fuzzy Priority Routing for K-TRP

Require: Customer set $C = \{c_1, c_2, \ldots, c_n\}$, depot D, cluster range $K_{\min}, K_{\max}$
Ensure: Routes $R = \{R_1, \ldots, R_K\}$, Total Latency
Load customer dataset
Extract features: (LocationX, LocationY, DistanceFromDepot, PlaceType)
Normalize all features
Determine optimal number of clusters using Elbow Method
Apply Fuzzy C-Means clustering
for each customer c_i **do**
  Compute membership values μ_{ij} for all clusters
end for
Assign one repairman to each cluster
for each cluster C_j **do**
  CurrentNode $\leftarrow$ Depot
  Unvisited $\leftarrow$ Customers in C_j
  Route $R_j \leftarrow \emptyset$
  ClusterLatency $\leftarrow 0$
  TravelDistance $\leftarrow 0$
  while Unvisited is not empty **do**
    for each customer u in Unvisited **do**
      Calculate DistanceScore
      Calculate WaitingTimeScore
      Obtain MembershipScore
      Apply Mamdani Fuzzy Rules
      Compute ServicePriority(u)
    end for
    Select $u^* = \arg\max_{u \in \text{Unvisited}} \big(\text{ServicePriority}(u)\big)$
    Append u^* to route R_j
    Update TravelDistance
    Update ClusterLatency
    CurrentNode $\leftarrow u^*$
    Remove u^* from Unvisited
  end while
end for
TotalLatency $\leftarrow \sum_{j=1}^{K} \text{ClusterLatency}_j$
return Routes, TotalLatency

V. Results and Analysis

To evaluate the effectiveness of the proposed Fuzzy C-Means (FCM) Clustering with Fuzzy Priority Routing framework, experiments were conducted on a synthetic dataset containing 1,200 customer locations distributed across a two-dimensional service region. Each customer record contains spatial coordinates, distance from the depot, and a place-type attribute representing different service categories. A common depot and five repairmen ($k = 5$) were considered throughout all experiments.

The proposed method was compared against two baseline approaches: Pure Greedy Routing and K-Means Clustering combined with Greedy Routing. The Pure Greedy method assigns customers solely based on nearest-neighbor selection without any clustering, whereas the K-Means approach first partitions customers into disjoint clusters before applying greedy route construction within each cluster. The proposed FCM+Fuzzy framework extends this idea by introducing soft clustering through Fuzzy C-Means and intelligent route selection through a Mamdani Fuzzy Inference System.

Figures 2, 3, and 4 illustrate the routes generated by the three approaches. In the Pure Greedy approach, repairmen frequently traverse across different service regions, resulting in route overlaps and inefficient customer assignments. Although K-Means clustering reduces this issue by creating geographically compact clusters, customers located near cluster boundaries are still forced into a single region, occasionally leading to suboptimal assignments. The proposed FCM approach addresses this limitation by allowing customers to possess partial membership across multiple clusters, resulting in more natural service-region formation.

Table II summarizes the overall performance of the evaluated approaches. The proposed FCM+Fuzzy framework achieved the lowest total latency of 830.54 km, compared to 919.69 km for Pure Greedy Routing and 880.96 km for K-Means+Greedy Routing. This corresponds to a reduction of 89.15 km over the Pure Greedy approach and 50.42 km over the K-Means baseline. In

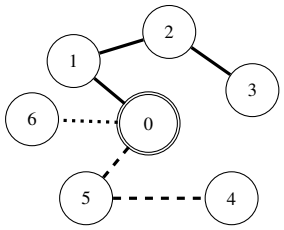


Fig. 2: Routes generated by Pure Greedy Routing. Repairmen cross service regions, causing overlaps.

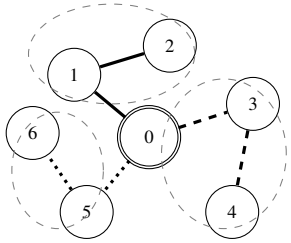


Fig. 3: Routes generated by K-Means + Greedy. Compact clusters reduce overlap but hard boundaries persist.

percentage terms, the proposed method achieved a latency reduction of 9.69% relative to Pure Greedy Routing and 5.72% relative to K-Means+Greedy Routing.

A similar trend can be observed in the average distance travelled per repairman. The proposed framework reduced the average travel distance to 166.11 km, compared to 183.94 km for Pure Greedy and 176.19 km for K-Means + Greedy. This indicates that the fuzzy routing strategy not only reduces overall latency but also distributes service responsibilities more efficiently among the available repairmen.

The observed improvements can be attributed to two key characteristics of the proposed framework. First, Fuzzy C-Means clustering generates soft cluster boundaries, allowing customers near regional borders to be assigned more appropriately based on membership values. Consequently, route fragmentation and unnecessary cross-region travel are reduced. Second, the Mamdani Fuzzy Inference System considers multiple decision factors simultaneously, including customer waiting time, distance from the current location, and cluster membership strength. Unlike traditional nearest-neighbor routing, which relies solely on geographic proximity, the proposed fuzzy routing mechanism produces more balanced and latency-aware service schedules.

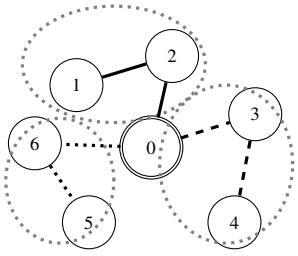


Fig. 4: Routes generated by FCM + Fuzzy Priority Routing. Soft boundaries enable more balanced assignments.

TABLE II: Performance Comparison of Routing Approaches

Metric	Pure Greedy	K-Means+Greedy	FCM+Fuzzy
Total Distance (km)	919.69	880.96	830.54
Average Distance (km)	183.94	176.19	166.11
vs. Pure Greedy (%)	—	4.21	9.69
vs. K-Means+Greedy (%)	—	—	5.72

Another important observation is the improved workload distribution among repairmen. In the Pure Greedy approach, some repairmen are assigned significantly longer routes than others due to uncontrolled customer selection. K-Means clustering partially alleviates this issue by partitioning the service region. However, the proposed FCM+Fuzzy framework further improves balance by considering customer-cluster affinity during route generation, resulting in more equitable service allocation across repairmen.

Overall, the experimental results demonstrate that the integration of fuzzy clustering and fuzzy decision-making provides a practical and computationally efficient solution for large-scale latency minimization problems. The consistent reduction in total latency and average travel distance confirms the effectiveness of the proposed framework and highlights its suitability for smart-city service systems, urban maintenance operations, and other real-world field service applications where minimizing customer waiting time is the primary objective.

VI. Application: Smart Urban Field Service Optimization

To demonstrate the practical applicability of the proposed FCM + Fuzzy Priority Routing framework, we consider the problem of smart urban field service optimization. In modern cities, service providers such as utility maintenance teams, telecommunication technicians, appliance repair services, and emergency response units must serve a large number of geographically distributed customer requests while maintaining low response times. In such environments, minimizing customer waiting time is often more important than minimizing travel distance alone.

A. Problem Mapping

The urban field service system can be naturally modeled as a K-Traveling Repairman Problem as follows:

- **Depot:** Central service office or dispatch center.
- **Customers:** Service request locations distributed across the city.
- **Repairmen:** Field technicians or service vehicles.

- **Travel Cost:** Estimated travel time or geographic distance between locations.
- **Objective:** Minimize cumulative customer waiting time (latency).

Each customer request must be serviced exactly once, and customer satisfaction is strongly influenced by response time. Therefore, reducing overall latency becomes the primary optimization objective.

B. Benefits of Fuzzy Clustering

In real-world urban environments, service requests rarely form perfectly separated geographic regions. Customers located near the boundaries of service zones may naturally belong to multiple operational regions. Traditional clustering approaches such as K-Means force every customer into a single cluster, potentially creating inefficient assignments.

The proposed Fuzzy C-Means (FCM) clustering approach addresses this limitation by assigning a degree of membership to every customer across multiple clusters. This allows the system to explicitly model geographic uncertainty and better represent boundary customers. Consequently, repairmen can operate within more coherent service regions while reducing route fragmentation and workload imbalance.

C. Fuzzy Priority-Based Service Scheduling

After clustering, routing decisions are generated using a Mamdani Fuzzy Inference System. Unlike conventional nearest-neighbor routing, which considers only distance, the proposed framework evaluates multiple decision factors simultaneously:

- Distance between the repairman and customer.
- Customer membership strength within the assigned cluster.
- Customer waiting time.

Using fuzzy IF–THEN rules, the system prioritizes customers who are geographically close, strongly associated with the current service region, or experiencing excessive waiting times. This multi-criteria decision process enables more balanced and customer-centric routing behavior.

D. Deployment Scenario

Consider a smart-city maintenance platform managing 1,200 service requests across multiple urban regions. Customer locations, service categories, and geographic coordinates are collected through a centralized management system. The proposed framework first applies Fuzzy C-Means clustering to partition the city into soft service regions and assign repairmen accordingly.

During route construction, the fuzzy priority engine continuously evaluates candidate customers and determines the most appropriate next visit based on distance, membership strength, and accumulated waiting time. Experimental results demonstrate that this approach reduces total system latency while simultaneously improving workload balance among repairmen.

The proposed framework is therefore well suited for large-scale smart-city applications including utility maintenance, telecommunication fault repair, healthcare home visits, municipal service operations, and emergency response logistics, where timely service delivery is critical for operational efficiency and customer satisfaction.

VII. Conclusion and Future Works

This paper presented a hybrid Fuzzy C-Means (FCM) and Fuzzy Priority Routing framework for solving the K-Traveling Repairman Problem (K-TRP), a latency-oriented routing problem in which multiple repairmen must collectively serve a set of customers while minimizing cumulative waiting time. Unlike traditional approaches that rely on hard clustering and deterministic nearest-neighbor routing, the proposed methodology incorporates soft clustering through FCM and a Mamdani fuzzy inference system that simultaneously considers distance, cluster membership strength, and customer waiting time when selecting the next customer to visit. This combination enables the routing process to better handle customer ambiguity near cluster boundaries while dynamically prioritizing customers based on multiple decision factors.

Experimental evaluation was conducted on a synthetic dataset containing 1,200 customers and five repairmen. The proposed FCM + Fuzzy Priority Routing approach was compared against two baseline methods: Pure Greedy Routing and K-Means + Greedy Routing. The results demonstrated that the proposed framework consistently achieved superior performance, reducing total system latency by 9.69% compared to the Pure Greedy approach and by 5.72% compared to the K-Means + Greedy approach. In addition to latency reduction, the fuzzy framework produced more balanced route distributions among repairmen, indicating improved workload equity and enhanced utilization of available service resources.

The findings of this study demonstrate that integrating fuzzy clustering and fuzzy decision-making into latency-based routing systems can significantly improve both operational efficiency and service quality. By replacing rigid assignment and routing decisions with soft, interpretable reasoning mechanisms, the proposed approach offers a scalable and computationally efficient alternative for large-scale service routing applications. Furthermore, the modular nature of the framework allows the clustering and routing components to be extended independently, making it suitable for a wide range of logistics, maintenance, and smart-city service scenarios.

Although the proposed framework produces promising results, several opportunities remain for future research. Future work may incorporate dynamic customer arrivals

and real-time route updates, enabling the system to operate in online environments where service requests appear continuously. The integration of real road-network distances, traffic-aware travel times, and heterogeneous repairmen with varying capabilities would further improve practical applicability. Additionally, adaptive fuzzy systems, neuro-fuzzy models such as ANFIS, and machine-learning-based rule generation techniques could be explored to automatically learn routing policies from historical operational data. Evaluating the proposed framework on established benchmark datasets and comparing it against advanced metaheuristic approaches such as Genetic Algorithms, GRASP, and Variable Neighborhood Search also represents an important direction for future investigation.

REFERENCES

[1] Bruni, M. E., Khodaparasti, S., Martínez-Salazar, I., & Nucamendi-Guillén, S. (2022). The Multi-Depot k-Traveling Repairman Problem. *Computers & Operations Research*, 145, 105862. https://doi.org/10.1016/j.cor.2022.105862

[2] Tulabandhula, T., Rudin, C., & Jaillet, P. (2011). Machine Learning and the Traveling Repairman. In *Proceedings of the 28th International Conference on Machine Learning (ICML)*. ACM.

[3] Jothi, R., & Raghavachari, B. (2007). Approximating the k-Traveling Repairman Problem with Repair Times. *Journal of Discrete Algorithms*, 5(2), 293–303. https://doi.org/10.1016/j.jda.2006.05.001

[4] Ren, Z., He, L., Liu, H., & Zhou, Y. (2021). An Effective Hybrid Search Algorithm for the Multiple Traveling Repairman Problem with Profits. *Information Sciences*, 559, 107–127. https://doi.org/10.1016/j.ins.2021.01.032

[5] Harrelson, C. (2003). The k-Traveling Repairman Problem. Unpublished manuscript. MIT Laboratory for Computer Science.

[6] Blum, A., Chalasani, P., Coppersmith, D., Pulleyblank, W., Raghavan, P., & Sudan, M. (1994). The Minimum Latency Problem. In *Proceedings of the 26th Annual ACM Symposium on Theory of Computing (STOC)*, pp. 163–171. https://doi.org/10.1145/195058.195125

[7] Ban, O. (2017). A GRASP+VND Algorithm for the Multiple Traveling Repairmen Problem with Distance Constraints. *Informatica*, 28(3), 427–445. https://doi.org/10.15388/Informatica.2017.136

[8] Hmayer, A., & Ome Ezzine, I. (2013). A CLARANS-Based Heuristic Approach for the K-TRP. In *Proceedings of the International Conference on Metaheuristics and Nature Inspired Computing (META)*. IEEE.

[9] Nucamendi-Guillén, S., Martínez-Salazar, I., Angel-Bello, F., & Moreno-Vega, J. M. (2016). A Mixed Integer Formulation and an Efficient Metaheuristic Procedure for the k-Traveling Repairman Problem. *Journal of the Operational Research Society*, 67(8), 1121–1134. https://doi.org/10.1057/jors.2016.3

[10] Onder, G., Kara, I., & Derya, T. (2017). New Integer Programming Formulations for the Multiple Traveling Repairmen Problem. *AIP Conference Proceedings*, 1836(1). https://doi.org/10.1063/1.4981974

[11] Afrati, F., Cosmadakis, S., Papadimitriou, C. H., Papageorgiou, G., & Papakostantinou, N. (1992). The Complexity of the Travelling Repairman Problem. *RAIRO Theoretical Informatics and Applications*, 26(3), 209–223.

[12] Lu, X., & Wen, J. (2017). An Asymptotically Optimal Algorithm for the Dynamic Traveling Repairman Problem. In *Proceedings of the 56th IEEE Conference on Decision and Control (CDC)*, pp. 6677–6682. https://doi.org/10.1109/CDC.2017.8264669

[13] Bezdek, J. C., Ehrlich, R., & Full, W. (1984). FCM: The Fuzzy c-Means Clustering Algorithm. *Computers & Geosciences*, 10(2–3), 191–203. https://doi.org/10.1016/0098-3004(84)90020-7

[14] Mamdani, E. H., & Assilian, S. (1975). An Experiment in Linguistic Synthesis with a Fuzzy Logic Controller. *International Journal of Man-Machine Studies*, 7(1), 1–13. https://doi.org/10.1016/S0020-7373(75)80002-2

[15] Zadeh, L. A. (1965). Fuzzy Sets. *Information and Control*, 8(3), 338–353. https://doi.org/10.1016/S0019-9958(65)90241-X

[16] Toth, P., & Vigo, D. (Eds.). (2014). *Vehicle Routing: Problems, Methods, and Applications* (2nd ed.). SIAM. https://doi.org/10.1137/1.9781611973594

[17] Laporte, G. (1992). The Traveling Salesman Problem: An Overview of Exact and Approximate Algorithms. *European Journal of Operational Research*, 59(2), 231–247. https://doi.org/10.1016/0377-2217(92)90138-Y

[18] Scikit-fuzzy Development Team. (2023). scikit-fuzzy: Fuzzy logic toolbox for SciPy. Version 0.4.2. GitHub Repository. https://github.com/scikit-fuzzy/scikit-fuzzy